Exoplanet Atmospheres Problem Set - Solutions

Section I: Life & Habitability

1a. Calculation

Using the given formula, the habitable zone boundaries for water are calculated with an albedo of **$A = 0.3$**, stellar luminosity **$L_\star = 3.827 \times 10^{26}$ W**, and the Stefan-Boltzmann constant **$\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}**. The freezing (273 K) and boiling (373 K) temperature limits result in a habitable zone approximately between **0.73 AU and 1.37 AU**.

1b. Interpretation

This simplified model underestimates the habitable zone because it doesn't account for crucial factors. Atmospheric greenhouse effects and cloud cover, for instance, can extend the habitable zone by trapping heat and raising the surface temperature, allowing a planet to be habitable even at a greater distance from its star.

Section II: Interpreting Atmospheric Absorption Spectra

The primary features in an atmospheric absorption spectrum include:

- **H_2O**: Strong absorption bands around **0.94 µm, 1.13 µm, 1.4 µm, 1.9 µm, 2.7 µm, and 6.3 µm**.
- **CO_2**: Prominent bands around **2.0 µm, 4.3 µm, and 15 µm**.
- **O_2**: Noteworthy bands at **0.76 µm (the A-band) and 1.27 µm**.
- **O_3**: The Hartley band in the UV region at **0.25 µm and 9.6 µm**.
- **CH_4**: Significant absorption features around **1.65 µm, 2.3 µm, and 3.3 µm**.

Section III: Characterizing Atmospheric Loss

3a. Calculation

The escape velocity for **Terra II** is calculated to be **15.81 km/s**.

3b. Interpretation

Despite its higher escape velocity of 15.81 km/s compared to Earth, Terra II is unlikely to retain its atmosphere. It receives **100 times more XUV radiation** from its host star than Earth does from the Sun. According to the **Cosmic Shoreline model**, this high XUV flux would accelerate atmospheric loss. Therefore, Terra II would likely lose its atmosphere despite its strong gravitational pull.